BurstLink: Techniques for Energy-Efficient Video Display for Conventional and Virtual Reality Systems

Analysis by

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Motivation

The most common application in mobile systems is conventional planar video streaming.

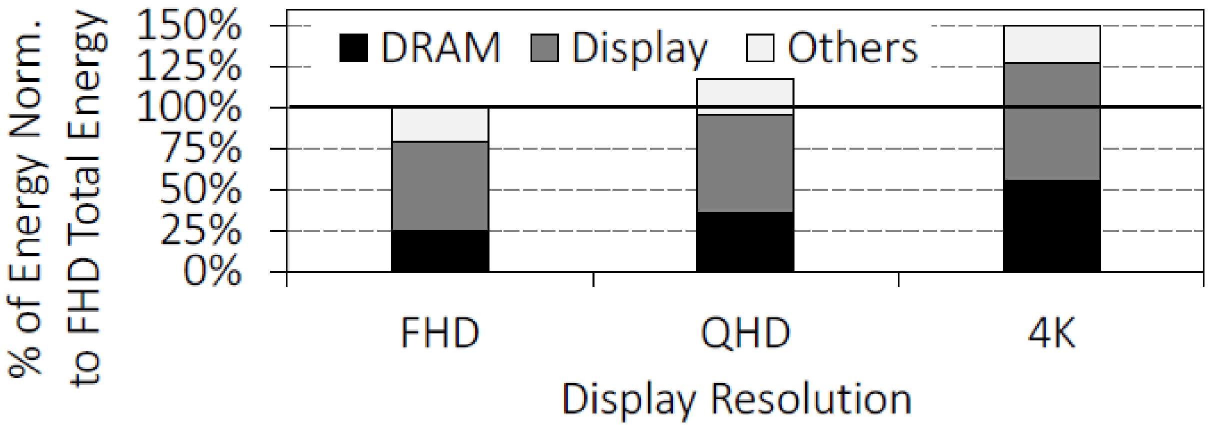
Cisco predicts that video streaming will generate more than 79% of mobile data traffic by 2022 (at time of publication).

The increasing proliferation of 360 video content and virtual reality (VR) devices is hastening VR video streaming adoption.

Problem to Face

Video formats and mobile display panels allow progressively high resolutions (e.g., 4K) and refresh rates (e.g., 120Hz) to provide consumers with an immersive experience.

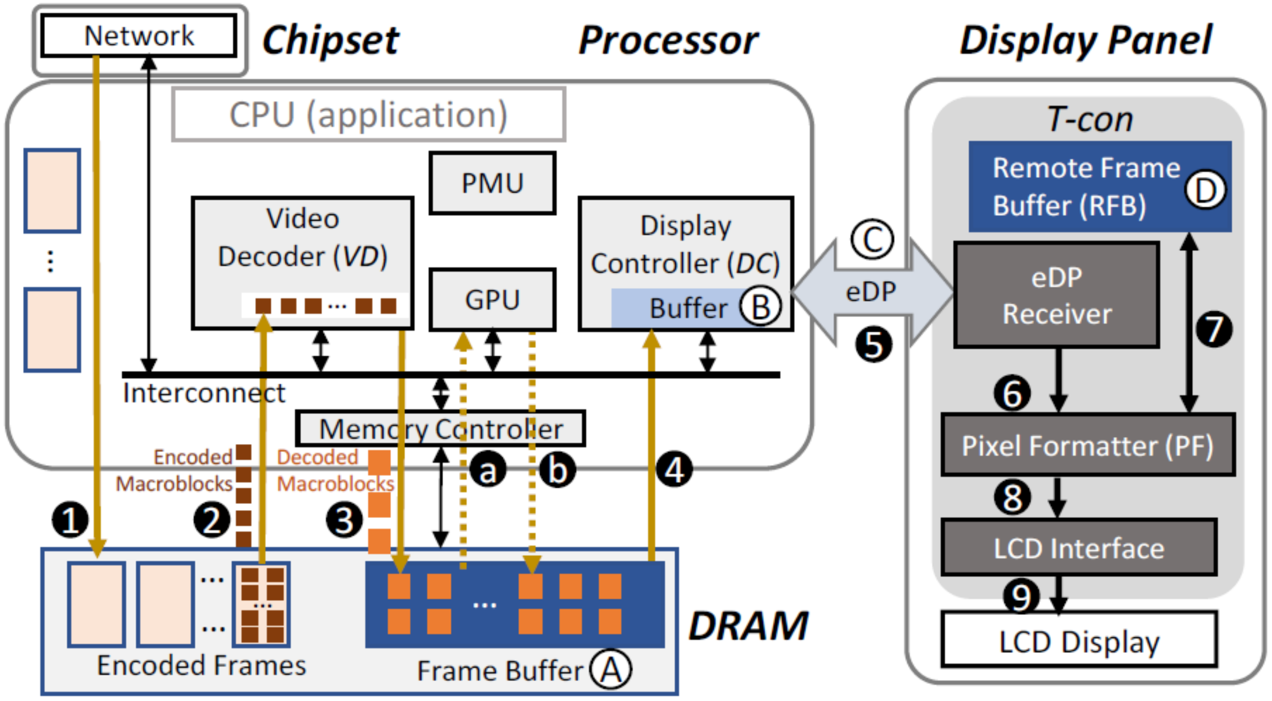
These trends come at the expense of dramatically increased video display energy usage.



Current System Architecture

- Inefficient in terms of Energy Consumption

- Does not utilize the total available bandwidth of embedded-Display Port (25.92 Gbps)



C-States

Package C-state	Major conditions to enter the package C-state
C0	One or more cores or graphics engine **executing instructions**
C2	All cores in **CC3** (clocks off) **or deeper** and graphics engine in **RC6** (power-gated). DRAM is **active**.
C3	All cores in **CC3 or deeper** and graphics engine in **RC6**. Last-Level-Cache (LLC) may be flushed and turned off, DRAM in **self-refresh (SR)**, most IO and memory domain clocks are gated, some IPs and IOs can be **active** (e.g., **DC and Display IO**).
C6	All cores in **CC6** (power-gated) or **deeper** and graphics engine in **RC6**. LLC may be flushed and turned off, DRAM in **self-refresh**, IO and memory domain clock generators are turned off. Some IPs and IOs can be **active** (e.g., **VD, DC**).
C7	Same as Package C6 while some of the IO and memory domains are **power-gated**.
C8	Same as Package C7 with additional **power-gating** in the IO and memory domains. **Only DC and Display IO are ON**.
C9	Same as Package C8 while all IPs must be off. Most VRs voltage are reduced. **The display panel can be in PSR**.
C10	**Same as Package C9 while all SoC VRs** (except always-on VR) **are off. The display panel is off**.

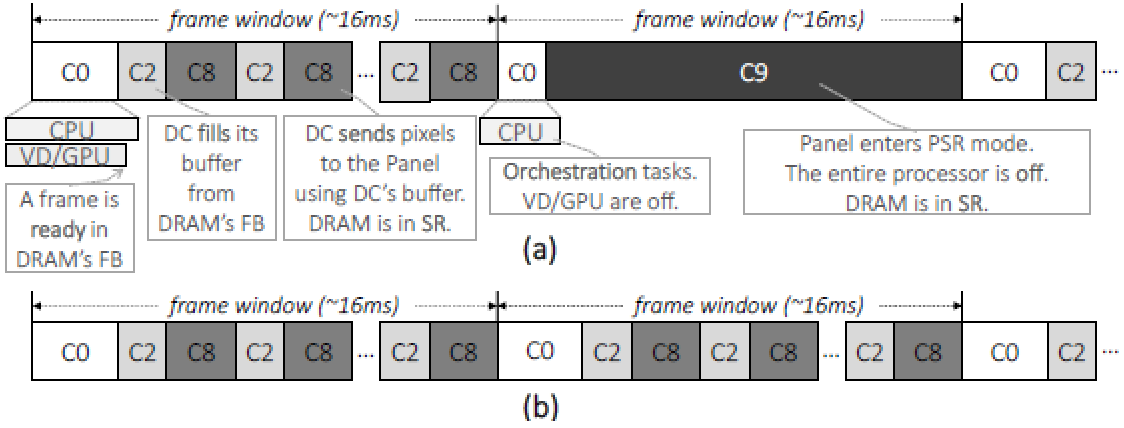


Figure 3: Package C-state timeline while displaying (a) a 30FPS video and (b) a 60FPS video on a 60Hz display panel.

Frame Buffer Bypassing

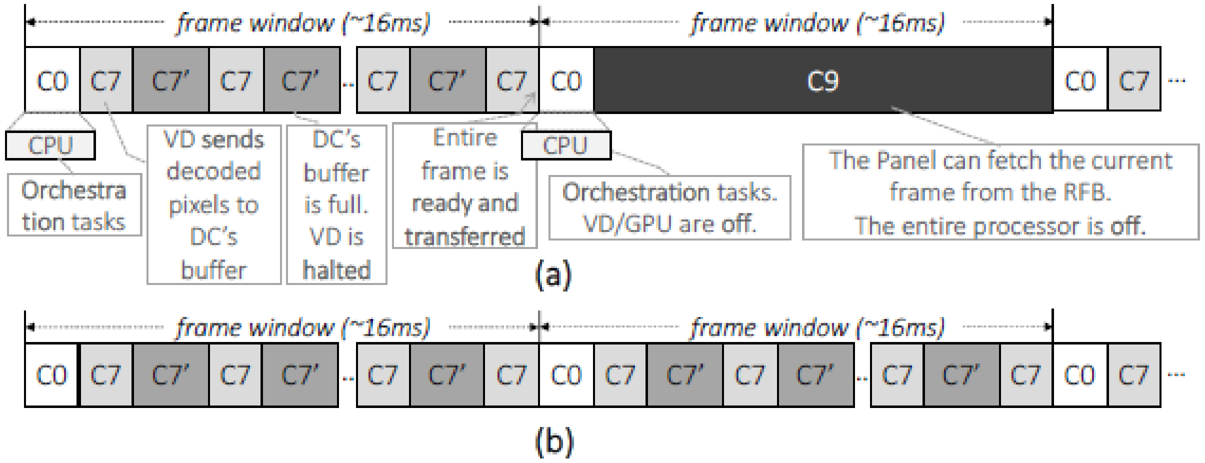


Figure 6: Package C-state timeline of a mobile processor using the Frame Buffer Bypass technique while displaying (a) a 30FPS video and (b) a 60FPS video on a 60Hz display panel.

Frame Bursting

A method of sending the decoded frame from the processor to the display panel in a burst.

The Frame Bursting technique involves the display panel receiving a full-frame across the embedded Display Port interface and directly storing it in the Double Remote Frame Buffer.

BurstLink – Final Architecture

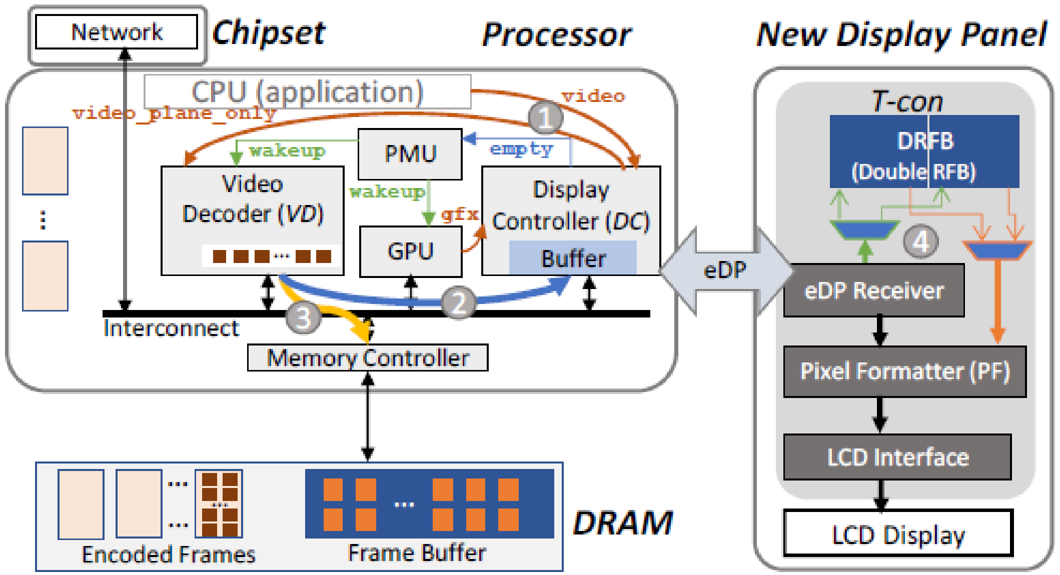


Figure 5: BurstLink video processing and display.

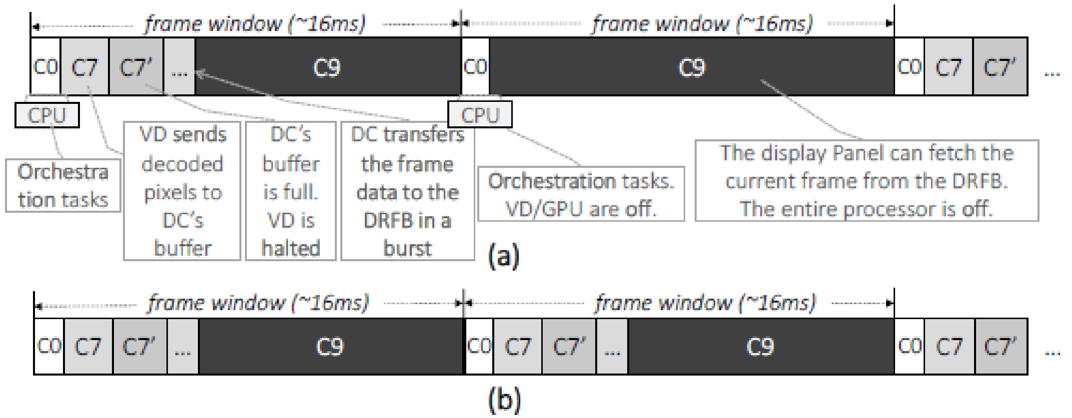
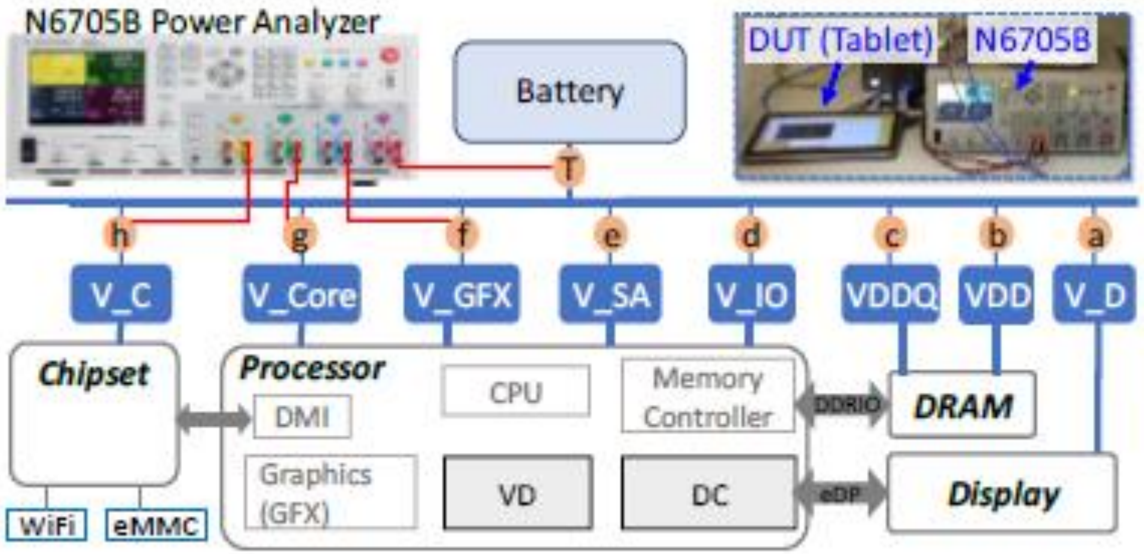
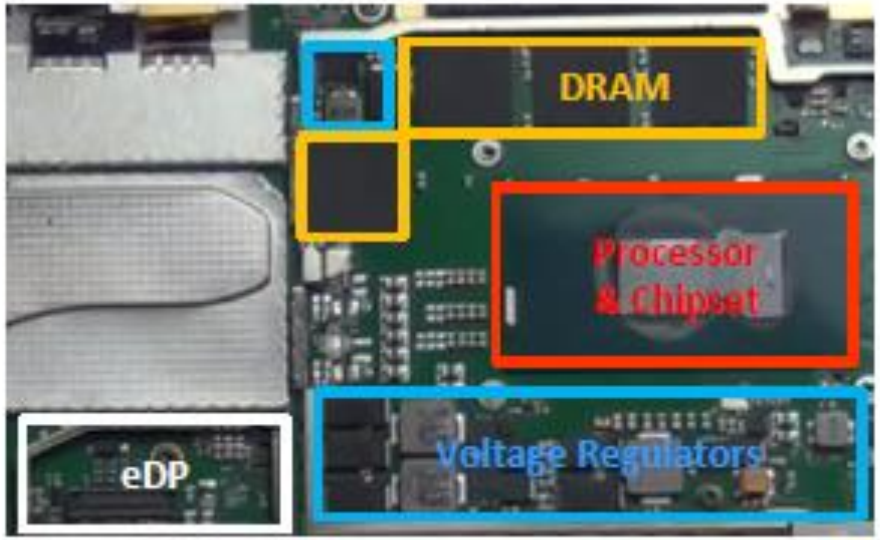


Figure 7: Package C-state timeline of a mobile processor with complete BurstLink for (a) 30FPS and (b) 60FPS videos on a 60Hz display panel.

Methodology





Conclusion

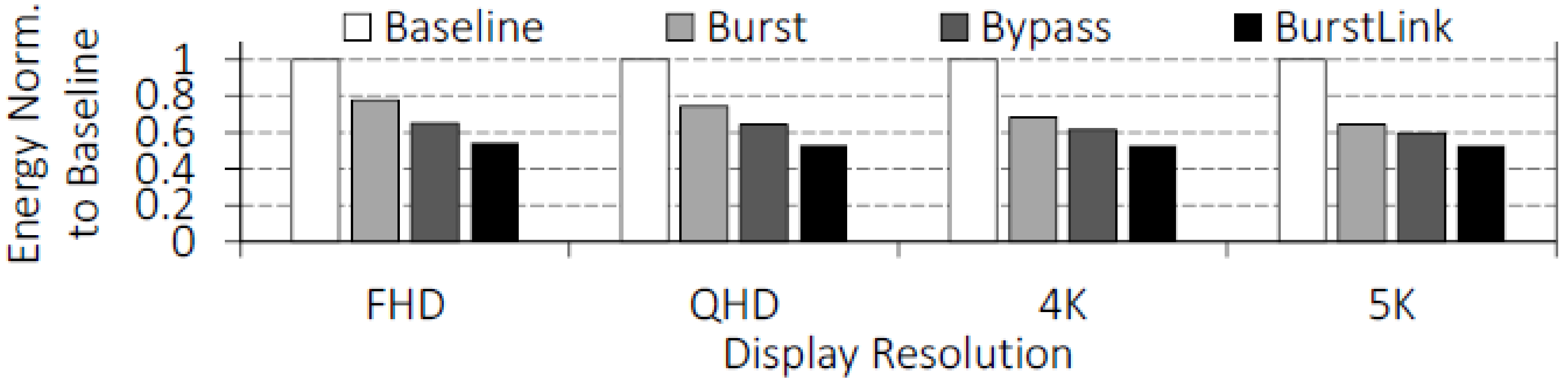


Figure 12: Total system energy reduction of Frame Bursting, Frame Buffer Bypassing, and BurstLink for 60 FPS HD videos.